Beyond the Shortcut: Medical Specialty Classification with and without Label-Revealing Keywords

Group 54: Catalina Trajterman (33.3%), Mathilde Saragoussi (33.3%) , Maria Oliveira Maia (33.3%)

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1 Introduction

Medical specialty classification consists in assigning clinical notes to the appropriate medical specialty based on their textual content. However, the presence of keywords can make this task misleadingly easy. In our dataset, the first keyword directly reveals the specialty label whenever keywords are available, allowing a model to bypass the actual medical text.

This raises our research question: *How well can models classify medical notes when keywords are missing, and how does their performance compare to classification using keywords?* Approximately 20% of the test notes have no keywords, making this a significant challenge.

2 Data

2.1 Data Analysis

The training set has 2,669 notes (description, sample name, transcription, keywords) labelled with 12 specialties; the test set has 398 unlabelled notes. The median transcription has 386 words. The raw training file contains malformed records (unbalanced quotes, line breaks inside records), so a standard CSV reader silently loses 53 training records; so we wrote a dedicated parser. We highlight three observations.

1-Strong class imbalance. Surgery holds 893 notes (33.5%), whereas Dermatology holds 25 (0.9%), a ratio of 36:1 (Fig. 1). A classifier that always predicts Surgery obtains 33.5% accuracy, so we have to report macro-F1 and the F1 of each class.

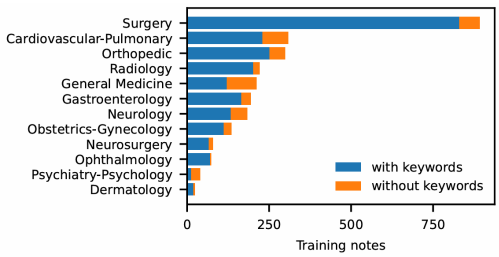


Figure 1: Training notes per specialty and in orange, the amount of those notes without keywords.

2-Keywords reveal the label, but are missing unevenly. In all 2,217 notes that have keywords (83.1%), the first keyword is exactly the specialty name. The rule "first keyword = label" is in fact always correct on those notes. The difficulty lies in the remaining notes, and these are concentrated in a few classes: 68% of Psychiatry-Psychology and 42% of General Medicine notes have no keywords, against 7% for Surgery and 4% for Ophthalmology (Fig. 1). Only 80% of the test notes have keywords, so validation scores may be slightly optimistic compared to the test evaluation.

3-The same note appears under several specialties. Among the 2,638 notes with a transcription, 1,534 (58%) share their exact text with at least one other note, and 731 of the 732 repeated texts carry different labels. Moreover, 239 of the 398 test notes (60%) also appear in the training set; since copies almost never share a label, the label of the training copy is probably not the label of the test note.

2.2 Data Augmentation

Random Deletion and Random Swap were evaluated separately, modifying only the description and transcription fields while preserving labels, keywords, and other attributes. Each method generated two variants per training note; validation notes were not augmented.

Random Deletion. Each word was removed with probability 0.05. Texts shorter than ten words were unchanged. The original text was retained if fewer than half its words remained.

Random Swap. One randomly selected pair of words was swapped in each text containing at least ten words; shorter texts were unchanged. Both methods introduce small textual variations, but may alter clinically relevant information. Their relative performance was therefore evaluated empirically.

Although both techniques are computationally inexpensive and straightforward to implement, they may introduce noise or alter clinically relevant information. For example, removing a medically relevant term or changing the order of words may affect the meaning of a clinical text. Therefore, their effectiveness cannot be assumed and must be evaluated empirically by comparing classification performance across experiments using the original training data and data augmented with each method separately.

3 Models

We adopted an incremental modeling strategy, from Complement Naive Bayes to Logistic Regression and a weighted BiLSTM, to investigate whether increasing model complexity improves classification. Each approach was evaluated with and without `keywords` to assess reliance on label-revealing cues.

3.1 Complement Naive Bayes

The baseline is Complement Naive Bayes [1], a variant of Naive Bayes designed for imbalanced text data, which estimates each class's feature weights from the complement of that class. The input is the concatenation of `sample_name`, `description` and `transcription` (plus `keywords` when enabled), represented with TF-IDF over the 10,000 most fre-

quent unigrams. We use sublinear term frequency ($1 + \log \mathrm{tf}$), `min_df`$= 2$ (terms must occur in at least two documents) and additive smoothing $\alpha = 0.1$.

3.2 Logistic Regression

Next, we implemented Logistic Regression [2] as a discriminative linear classifier using TF-IDF text representations. The model uses both unigrams and bigrams to represent the textual features and combines the description and transcription fields, optionally including keywords. The TF-IDF transformation and Logistic Regression classifier are integrated into a single pipeline. To account for class imbalance, we used balanced class weights. As in the previous experiments, we evaluated configurations with and without `keywords`.

3.3 Weighted BiLSTM

Finally, we implemented a weighted Bidirectional Long Short-Term Memory (BiLSTM) [3] network to explore a neural approach capable of capturing sequential and contextual information. We incorporated class weights to address the strong class imbalance and give greater importance to underrepresented specialties during training. We again compared configurations with and without `keywords` to assess whether contextual representations improve classification and how explicit specialty-related information affects performance.

3.4 Comparison Strategy

Across all three approaches, keyword availability was varied independently of the model architecture. Comparing results with and without keywords helps assess whether models learn from clinical content or rely on explicit label-related cues. The comparison across Naive Bayes, Logistic Regression, and BiLSTM additionally allows us to investigate whether moving from lexical features to contextual representations improves classification when keywords are absent.

4 Experimental Setup

4.1 Datasets and Splits

Experiments were conducted on 2,669 labeled training notes covering 12 medical specialties and 398 unlabeled test notes. Inputs consisted of the `description`, `sample_name`, and `transcription` fields, with `keywords` optionally included. Models were evaluated with and without keywords to assess their contribution to classification performance.

For evaluation, the labeled data was split into training and validation sets using a 90/10 stratified split with a random seed of 42. The validation set was excluded from data augmentation to prevent data leakage. The unlabeled test set was reserved for final predictions. For augmentation experiments, Random Deletion and Random Swap were evaluated separately.

For each augmentation method, two variants were generated per original note, producing 7,206 records including the originals.

4.2 Evaluation Metrics

Performance was assessed using accuracy, precision, recall, and F1-score. Given the strong class imbalance, macro-averaged metrics were reported to give equal weight to each specialty, alongside per-class precision, recall, and F1-scores. Confusion matrices were also used to examine class-specific errors for the final model.

4.3 Hyperparameters

Naive Bayes and Logistic Regression used TF-IDF text features. Logistic Regression used unigrams and bigrams with balanced class weights. The BiLSTM used a vocabulary limited to 30,000 tokens, sequences truncated or padded to 1,000 tokens, and 128-dimensional embeddings. Its architecture consisted of a bidirectional LSTM with 64 units, followed by a 64-unit ReLU layer and dropout (0.3). The LSTM also used dropout of 0.2. The model was trained with Adam, sparse categorical cross-entropy, a batch size of 32, and a maximum of 5 epochs (when changed to 10 the accuracy did not improve significantly, approx. 1%). Early stopping monitored validation loss with a patience of 2 and restored the best weights.

5 Results

Table 1 reports validation results for both augmentation methods, with and without keywords. The selected model is the weighted BiLSTM with Random Deletion and keywords, which achieved 73.78% accuracy, a macro-F1 of 0.70, a weighted F1 of 0.74, and a validation loss of 0.8279. On the validation set, this model exceeded 49% accuracy and achieved a minimum per-class F1-score of 0.44, above the required threshold of 0.25, so it is the model used to generate `results.txt`.

Model	Aug.	Keywords	Acc.	Macro-F1
NB	del.	no	0.431	0.372
NB	del.	yes	0.464	0.411
LR	del.	no	0.441	0.49
LR	del.	yes	0.52	0.55
BiLSTM	del.	no	0.36	0.37
BiLSTM	del.	yes	0.738	0.70
NB	swap	no	0.431	0.376
NB	swap	yes	0.461	0.403
LR	swap	no	0.44	0.49
LR	swap	yes	0.52	0.56
BiLSTM	swap	no	0.401	0.40
BiLSTM	swap	yes	0.719	0.64

Table 1: Validation results (267 notes). NB: Complement Naive Bayes; LR: Logistic Regression; del.: Random Deletion; swap: Random Swap.

Keywords improved all three models under both augmentation methods, with the largest accuracy gains observed for the BiLSTM. Without keywords, Logistic Regression achieved the highest macro-F1 (0.49) under both methods, and no configuration exceeded 49% validation accuracy. For the runs reported in Table 1, the BiLSTM with keywords achieved 73.78% accuracy with Deletion and 71.91% with Swap.

For the selected BiLSTM with Random Deletion and keywords, Radiology and Neurosurgery achieved the highest per-class F1-scores (0.84), followed by Gastroenterology (0.80) and Orthopedic (0.79). Ophthalmology and Derma-

tology obtained the lowest scores, at 0.44 and 0.50, respectively. Neurosurgery achieved recall of 1.00 on eight validation notes; Dermatology achieved recall of 0.67 and precision of 0.40 on three notes. Figure 2 presents the confusion matrix for this model, with 197 correct predictions and 70 errors among 267 validation notes.

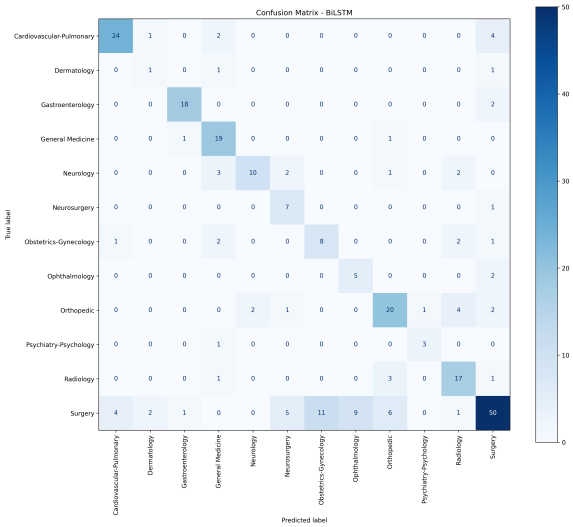


Figure 2: Confusion matrix of the BiLSTM with Random Deletion and keywords on the validation set.

6 Discussion

Augmentation comparison. Random Deletion and Random Swap produced similar results for the TF-IDF models. For Naive Bayes, Swap slightly increased macro-F1 without keywords (0.376 versus 0.372), but reduced it with keywords (0.403 versus 0.411). Logistic Regression obtained similar accuracy under both methods, with macro-F1 with keywords increasing slightly from 0.55 to 0.56.

For the runs reported in Table 1, Swap increased BiLSTM accuracy without keywords from approximately 36% to 40.07%. With keywords, Deletion achieved 73.78% accuracy and a macro-F1 of 0.70, compared with 71.91% and 0.64 for the saved Swap run. However, another Swap run achieved 74.16% accuracy, so these results do not establish consistent superiority of either method. Deletion was retained as the submission configuration. Without an unaugmented baseline, these experiments cannot establish whether either method improves on training with original notes alone.

Main errors. The selected BiLSTM with Random Deletion and keywords misclassifies 70 of the 267 validation notes, and Surgery is involved in 40 of them: 21 Surgery notes are predicted as another specialty (6 as Ophthalmology) and 19 notes from other specialties are predicted as Surgery (8 from Cardiovascular-Pulmonary). Surgery is related to a medical act, not a body area, so an eye or heart operation fits both labels. This is why Ophthalmology precision is so low (0.36): 6 of its 11 predictions are Surgery notes. General Medicine also collects errors (precision 0.61), mainly symptom-based Neurology and Cardiovascular-Pulmonary notes.

Examples (validation set, true → predicted):

1. Surgery → Ophthalmology (conf. 0.57): *"Cataract, right eye. Phacoemulsification with intraocular lens insertion [. . .] surgery, tetracaine, lid speculum [. . .]"*. It is an eye operation with about ten eye-related keywords, which outweigh the keyword "surgery". The prediction makes medical sense, so the label is ambiguous.

2. Surgery → Ophthalmology (conf. 0.93): *"Cervical facial rhytidectomy. Quadrilateral blepharoplasty [. . .] facelift [. . .] right upper lid"*. This is a facelift, but *blepharoplasty* and *upper lid* are typical of eye notes. The model seems to react to single words rather than the whole procedure.

3. Cardiovascular-Pulmonary → Surgery, no keywords (conf. 0.99): *"Implantation of a single-chamber pacemaker [. . .] chronic atrial fibrillation [. . .] Ancef 1 g [. . .] Wound check in 7 to 10 days."* The content is cardiac, but the note is written like an operative report, and the model follows that structure. Seven of the 8 notes of this class predicted as Surgery are procedure reports.

Research question. 38 of the 70 errors are on notes with keywords and 32 on notes without. Since only 16.5% of validation notes lack keywords, the error rate is about 73% without them against 17% with them, so notes without keywords are much harder. Keywords are not a guaranteed shortcut for the model: in 32 of the 38 errors the true label appears in the input and is still ignored. When trained and evaluated without the keywords field, the BiLSTM achieved approximately 36% accuracy and a macro-F1 of 0.37.

7 Future Work

Future work could evaluate paraphrasing and back-translation, checking that medical terminology and clinical meaning are preserved. Fine-tuning BioBERT or ClinicalBERT could also assess whether pre-trained representations improve classification without keywords. These experiments were not conducted due to computational constraints.

Bibliography

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AI Assistance Declaration. LLMs assisted with code debugging, methodological clarification, and report drafting. Training and evaluation were performed through the project scripts. The authors are responsible for verifying the code, results, and interpretations.